



NexusHR – AI-Enabled Enterprise HR & Workforce Intelligence Platform

Employee Lifecycle • Attendance • Payroll • Performance • AI
Insights

Production-Grade Java Full-Stack System Zidio Development / Portfolio / Interview

Reference May 2026



Title: NexusHR – AI-Enabled Enterprise HR & Workforce Intelligence Platform

subtitle: Production-Grade Java Full-Stack HR System with Real-Time Analytics & AI Insights

Author: Zidio Development ✨

Prepared-for: Zidio Development – Java Full-Stack Domain

Date: March 2026

version: 2.0 – Industry Edition

"HR teams spend too much time on manual processes and lack real-time workforce insights. NexusHR streamlines employee lifecycle, attendance, payroll, performance tracking, and delivers AI-powered insights – reducing administrative workload by 40–60% and improving decision-making."

1. Business Case & Production Objectives

Mission

Build a production-grade Java full-stack HR management platform that covers the complete employee lifecycle – from onboarding to offboarding – with real-time attendance, automated payroll, performance reviews, and AI-driven workforce intelligence. The system is designed for mid-to-large enterprises to reduce HR operational costs and improve employee experience.

Why This Project Matters for Zidio Development

Modern enterprises need integrated HR systems that eliminate spreadsheets and manual tracking. NexusHR provides a scalable, secure, and intelligent solution that Zidio can offer to clients in manufacturing, IT services, banking, and retail sectors.

Quantified Business Impact Targets

- Reduce HR administrative workload by **40–60%** through automation
- Improve employee retention by **15–25%** via better performance insights and engagement tracking
- Support **5,000–50,000 employees** per deployment
- Achieve **99.95%** uptime SLA
- Zero-downtime deployments for new features

Non-Functional Requirements

- Latency: < 300 ms for core API calls, < 2 s for dashboard loads
- Throughput: Handle 10k+ concurrent HR users during payroll cycles
- Availability: 99.95%
- Security: OWASP Top 10 mitigation, role-based access, data encryption at rest
- Scalability: Horizontal auto-scaling with Kubernetes
- Observability: Full logging, metrics, and audit trails

2. Core Functional Requirements

ID	Capability	Detailed Description & Business Value	Key Acceptance Criteria & Production Metrics
F-01	Employee Lifecycle Management	Onboarding, profile management, role assignment, offboarding	Complete workflow with approval steps, document upload
F-02	Attendance & Leave Management	Biometric simulation, leave requests, approval workflow	Real-time attendance dashboard, accurate leave balance calculation
F-03	Payroll Processing	Automated salary calculation, tax deduction, payslip generation	Accurate payroll runs, exportable payslips
F-04	Performance Management	Goal setting, reviews, 360-degree feedback, rating system	Performance scorecards, trend analysis
F-05	AI Workforce Insights	Predictive attrition, skill gap analysis, engagement scoring	AI recommendations with >80% accuracy on sample data
F-06	Admin & Manager Dashboards	Role-based views with analytics and reporting	Real-time metrics, export options (PDF/Excel)
F-07	Notification & Communication	Email/SMS alerts for approvals, reminders, announcements	Multi-channel delivery with 95% success rate

3. Production Technology Stack – 2026

Layer	Primary Technology	Rationale / Alternatives
Backend	Java 21 + Spring Boot 3.3	Enterprise standard, virtual threads, excellent ecosystem
Frontend	React 19 + TypeScript + Vite	Fast development, modern UI
UI Components	shadcn/ui + Tailwind CSS v4	Consistent, accessible design
Database	PostgreSQL 17	ACID compliance, JSONB for flexible data

Layer	Primary Technology	Rationale / Alternatives
Cache	Redis 7+	Session management, real-time data
Authentication	Spring Security 6 + JWT	Secure, role-based access control
AI Integration	Spring AI + OpenAI / Hugging Face	Intelligent insights and recommendations
Containerization	Docker multi-stage	Consistent, lightweight images
Orchestration	Kubernetes + Helm	Auto-scaling, production deployment
CI/CD	GitHub Actions	Automated builds, tests, and deployments
Monitoring	Prometheus + Grafana + Sentry	Full observability and alerting

4. 28-Days Day-by-Day Execution Plan

Week 1 – Core Backend & Authentication Foundation

Day 1

- Maven multi-module project setup (api-gateway, auth-service, employee-service, payroll-service, common)
- Spring Boot 3.3 initializer with required starters
- Basic server configuration and initial git commit

Day 2

- Employee and Role entities with JPA/Hibernate
- Spring Security configuration with JWT filter and Argon2 password encoder

Day 3

- Authentication endpoints (signup, login, refresh, logout)
- Role-based access control using @PreAuthorize

Day 4

- PostgreSQL schema with Flyway migrations for employees, roles, departments
- Basic employee CRUD operations

Day 5

- Redis integration for session caching
- Attendance tracking module foundation

Day 6

- Leave management service with approval workflow

Day 7

- Week 1 checkpoint: Backend API running, authentication working, basic employee and attendance modules tested with Postman

Week 2 – Payroll, Performance & Frontend Foundation

Day 8

- React 19 app bootstrap with TypeScript, shadcn/ui, Tailwind, TanStack Query

Day 9

- Authentication pages and protected routes in frontend

Day 10

- Payroll calculation logic and payslip generation

Day 11

- Performance review module with rating system

Day 12

- Employee dashboard UI with key metrics

Day 13

- Manager dashboard with team overview

Day 14

- Week 2 checkpoint: Payroll and performance modules functional, frontend connected to backend

Week 3 – AI Insights & Real-Time Features

Day 15

- AI integration setup using Spring AI for attrition prediction

Day 16

- Skill gap analysis and engagement scoring logic

Day 17

- Real-time notification system using WebSocket

Day 18

- Admin dashboard with workforce analytics

Day 19

- Email/SMS notification service for approvals and reminders

Day 20

- Dark mode implementation across frontend

Day 21

- Week 3 checkpoint: AI insights and real-time notifications operational

Week 4 – Deployment, Monitoring & Production Polish

Day 22

- Docker multi-stage builds for backend and frontend

Day 23

- Kubernetes manifests with HPA for auto-scaling

Day 24

- GitHub Actions CI/CD pipeline with tests and deployment

Day 25

- Cloud deployment on AWS EKS with S3 for documents

Day 26

- Prometheus + Grafana monitoring setup

Day 27

- Load testing and security scan with OWASP ZAP

Day 28

- Final QA, README polishing, demo video recording, PDF export

5. Challenges, Learnings & Industry Best Practices

- Handling complex payroll rules and compliance
- Integrating AI models for actionable HR insights
- Best practices: Multi-module architecture, CI/CD, observability, RBAC

6. Security & Privacy Highlights

- JWT with refresh tokens
- OWASP Top 10 mitigation
- Data encryption at rest
- Role-based access control
- Audit logging for sensitive operations

Zidio Development – Java Full Stack Domain

Project Submission Guidelines

March 2026 Edition

Prepared for participants

Focus: Industry-grade full-stack / advanced web projects

Submission period: You need to submit the project on or before the due date if you submit after the due date means you will be automatically disqualified for stipend by the system.

Evaluation emphasis: Code quality · Documentation · Live demo · Scalability & security awareness

1. General Rules & Eligibility

- All code must be **original work** created mainly during the ZIDIO preparation/submission window.
- Participants must submit **exactly 1 project** (as per the planned structure).
- Projects should demonstrate **progressive complexity**:
 1. Strong modern frontend / real-time foundations
 2. Distributed systems / microservices / resilience
 3. AI/ML-integrated or data-heavy web application
- Plagiarism / Ai generated content will result in disqualification for stipend.

2. Mandatory Submission Deliverables

Submit **one consolidated package** containing all three projects:

#	Deliverable	Format / Location	Required?	Evaluation Weight
1	Project Documentation / Report (1 PDF)	1 PDF file (A4, 8–15 pages)	Yes	25%
2	Live Public Demo URL	1 HTTPS link (no mandatory login)	Yes	30%
3	GitHub Repository	1 repo with clear folders	Yes	20%
4	README.md per project	Detailed, professional README in repo	Yes	15%
5	Demo Video(s)	3–7 min per project (LinkedIn Post/ YouTube unlisted / Loom / Drive)	Yes	10%

Naming convention recommendation

NexusHR - AI-Enabled_Enterprise_HR_&_Workforce_Intelligence
PlatformZidio_March2026.pdf/zip

3. Documentation Guidelines – What every PDF should contain

Use consistent structure across all three documents for a polished portfolio feel.

1. Hero / Cover Section

- Large project title
- Catchy tagline
- Your name + date
- Gradient background (optional but recommended)

2. Project Overview

- Vision & objectives
- Target users / use cases
- Business value delivered
- Non-functional goals (latency, concurrency, availability targets)

3. Key Features (table format)

| ID | Feature | Description | Acceptance Criteria |

4. Technology Stack (table)

| Category | Technology | Rationale / Alternatives |

5. Architecture Diagram

- Include screenshot (Excalidraw, Draw.io, Lucidchart, etc.)
- Or clean ASCII art if no image tool used

6. Detailed Execution Timeline

- Day-by-day or week-by-week breakdown
- Major deliverables per phase
- Milestones & checkpoints

7. Technical Highlights

- Security measures (OWASP mitigation, input sanitization, rate limiting, etc.)
- Performance & scalability notes (caching, lazy loading, load test results)
- Challenges faced & how solved

8. Deployment & Operations

- Platform used (Vercel, Render, Railway, AWS, Fly.io, etc.)

- CI/CD pipeline summary
- Monitoring / health checks (if implemented)

9. Visuals

- 5–10 high-quality screenshots / GIFs
- Architecture diagram
- Live demo captures

10. Personal Reflection (optional but strongly recommended)

- Key learnings
- Industry best practices applied
- Future roadmap ideas

4. Code & Repository Expectations

- Clean, modular code structure
- Consistent naming & formatting (use ESLint / Prettier)
- Semantic commit messages (`feat: add real-time cursor presence` , `fix: JWT refresh token bug` , etc.)
- Feature branches & pull requests (even if solo)
- `.gitignore` properly configured
- No committed secrets / API keys
- Basic tests appreciated (even 30–50% coverage shows intent)
- Responsive design + accessibility basics (ARIA labels, keyboard navigation)

5. Live Demo Requirements

- Publicly accessible (no VPN / geo-restriction)
- HTTPS enforced
- Core functionality available without sign-up (demo account OK if needed)
- Fast initial load (< 5 seconds preferred)
- Mobile responsive (test on phone / tablet)
- Include brief instructions on demo page if non-obvious

6. Evaluation Criteria (Suggested 100-point scale)

Category	Points	Focus Areas
Innovation & Problem Solving	15	Original features, thoughtful design choices

Category	Points	Focus Areas
Technical Depth & Best Practices	25	Clean code, modern stack, security & performance awareness
Functionality & User Experience	20	Works as promised, intuitive UX, responsive
Documentation Quality	20	Clear, professional, well-structured
Deployment & Reliability	10	Stable live demo, easy local setup
Presentation & Polish	10	Demo video quality, visual appeal of docs

7. Important Rules & Tips

- **Strict deadline** – no late submissions accepted
- **File size limits:** ZIP $\leq$ 500 MB, individual PDFs $\leq$ 10 MB
- **No external dependencies that require payment / private keys** for judges to run demo
- **Backup plan:** If live demo is down, high-quality video + screenshots must still convey full functionality
- **Stand-out practices:**
 - Include Lighthouse scores / performance metrics
 - Show real multi-user session in video (especially Project 1)
 - Mention OWASP mitigations, load test results, CI/CD pipeline
 - End docs with personal growth reflection

Submission email / form subject example

Thers no email submission you need to submit the project in dashboard itself

Good luck!

Submit with confidence – your detailed planning and advanced topics already make these projects stand out.

Team Zidio Development

March 2026

